In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
dataset = pd.read_csv("UberDataset.csv")
dataset.head()
```

Out[3]:

	START_DATE	END_DATE	CATEGORY	START	STOP	MILES	PURPOSE
0	01-01-2016 21:11	01-01-2016 21:17	Business	Fort Pierce	Fort Pierce	5.1	Meal/Entertain
1	01-02-2016 01:25	01-02-2016 01:37	Business	Fort Pierce	Fort Pierce	5.0	NaN
2	01-02-2016 20:25	01-02-2016 20:38	Business	Fort Pierce	Fort Pierce	4.8	Errand/Supplies
3	01-05-2016 17:31	01-05-2016 17:45	Business	Fort Pierce	Fort Pierce	4.7	Meeting
4	01-06-2016 14:42	01-06-2016 15:49	Business	Fort Pierce	West Palm Beach	63.7	Customer Visit

In [4]:
```
dataset.shape
```

Out[4]:
```
(1156, 7)
```

In [5]:
```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1156 entries, 0 to 1155
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype  
---  ------      --------------  -----  
 0   START_DATE  1156 non-null   object 
 1   END_DATE    1155 non-null   object 
 2   CATEGORY    1155 non-null   object 
 3   START       1155 non-null   object 
 4   STOP        1155 non-null   object 
 5   MILES       1156 non-null   float64
 6   PURPOSE     653 non-null    object 
dtypes: float64(1), object(6)
memory usage: 63.3+ KB
```

In [6]:
```
dataset.isnull()
```

Out[6]:

	START_DATE	END_DATE	CATEGORY	START	STOP	MILES	PURPOSE
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	True
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...
1151	False	False	False	False	False	False	False
1152	False	False	False	False	False	False	False
1153	False	False	False	False	False	False	False
1154	False	False	False	False	False	False	False
1155	False	True	True	True	True	False	True

1156 rows × 7 columns

Transform Data

In [7]:
```
dataset['PURPOSE'].fillna("NOT", inplace=True)
```

In [8]:
```
dataset['START_DATE'] = pd.to_datetime(dataset['START_DATE'],
                                       errors='coerce')
dataset['END_DATE'] = pd.to_datetime(dataset['END_DATE'],
                                     errors='coerce')
```

In [9]:
```
from datetime import datetime

dataset['date'] = pd.DatetimeIndex(dataset['START_DATE']).date
dataset['time'] = pd.DatetimeIndex(dataset['START_DATE']).hour

#changing into categories of day and night
dataset['day-night'] = pd.cut(x=dataset['time'],
                              bins = [0,10,15,19,24],
                              labels = ['Morning','Afternoon','Evening','Night'])
```

In [10]:
```
dataset.dropna(inplace=True)
```

In [11]:
```
dataset.drop_duplicates(inplace=True)
```

In [12]:
```
obj = (dataset.dtypes == 'object')
object_cols = list(obj[obj].index)

unique_values = {}
for col in object_cols:
  unique_values[col] = dataset[col].unique().size
unique_values
```

Out[12]:
```
{'CATEGORY': 2, 'START': 175, 'STOP': 186, 'PURPOSE': 11, 'date': 291}
```

In [14]:
```
plt.figure(figsize=(10,5))
```

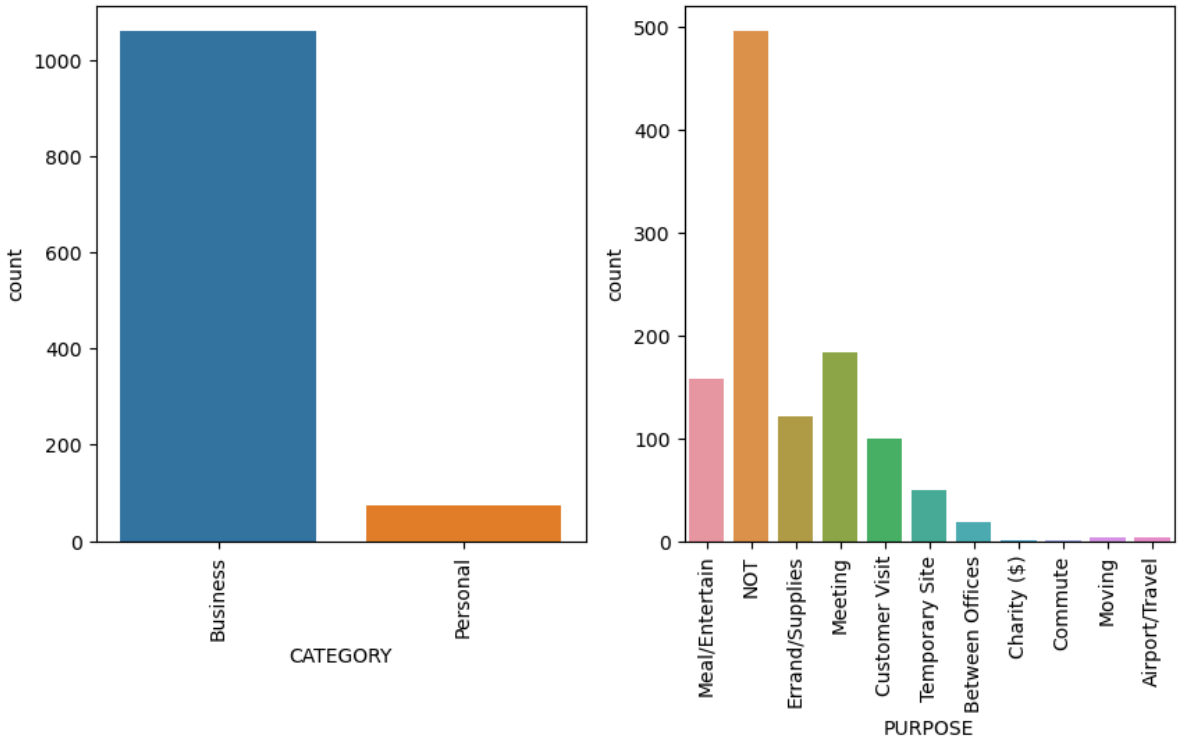
```
sns.countplot(dataset['CATEGORY'])
plt.xticks(rotation=90)

plt.subplot(1,2,2)
sns.countplot(dataset['PURPOSE'])
plt.xticks(rotation=90)
```

```
C:\Users\yashc\anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarnin
g: Pass the following variable as a keyword arg: x. From version 0.12, the only va
lid positional argument will be `data`, and passing other arguments without an exp
licit keyword will result in an error or misinterpretation.
  warnings.warn(
C:\Users\yashc\anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarnin
g: Pass the following variable as a keyword arg: x. From version 0.12, the only va
lid positional argument will be `data`, and passing other arguments without an exp
licit keyword will result in an error or misinterpretation.
  warnings.warn(
```

Out[14]:

```
(array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10]),
 [Text(0, 0, 'Meal/Entertain'),
  Text(1, 0, 'NOT'),
  Text(2, 0, 'Errand/Supplies'),
  Text(3, 0, 'Meeting'),
  Text(4, 0, 'Customer Visit'),
  Text(5, 0, 'Temporary Site'),
  Text(6, 0, 'Between Offices'),
  Text(7, 0, 'Charity ($)'),
  Text(8, 0, 'Commute'),
  Text(9, 0, 'Moving'),
  Text(10, 0, 'Airport/Travel')])
```

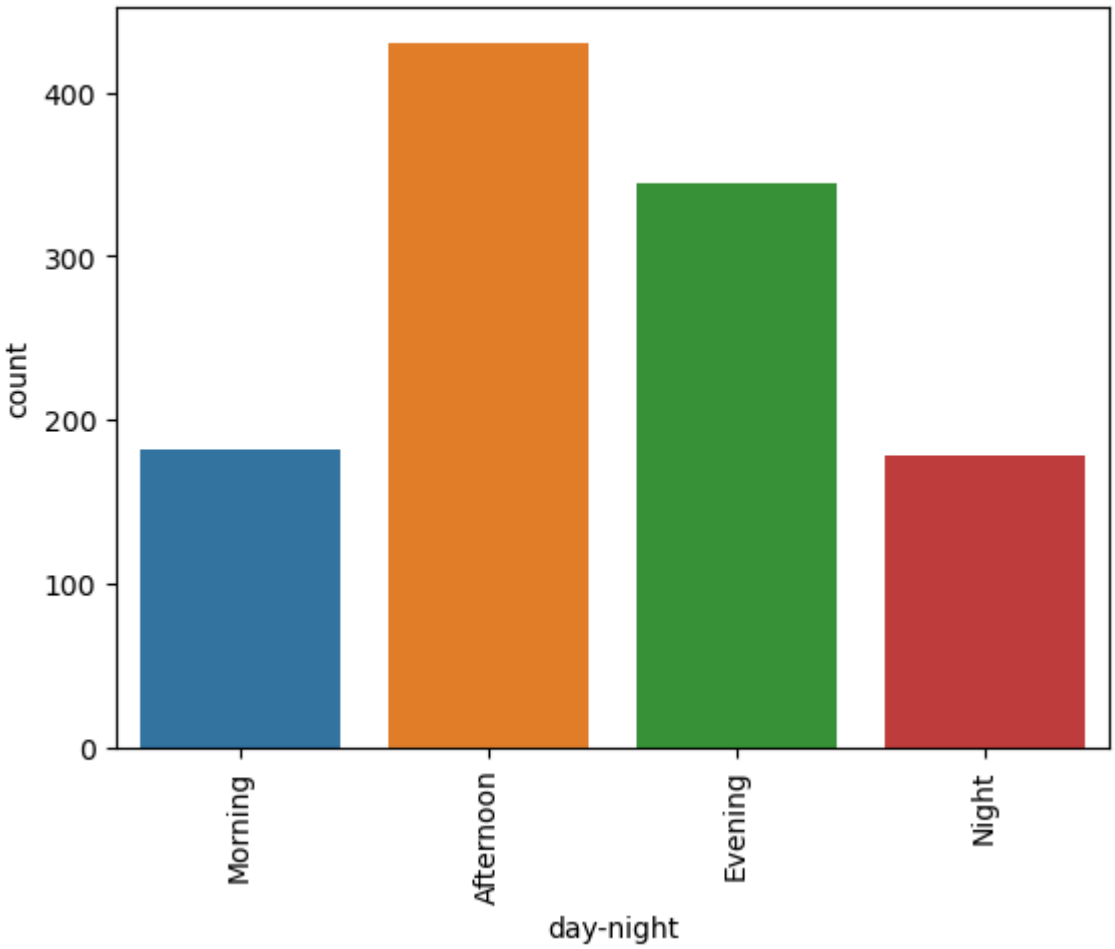


In [15]:

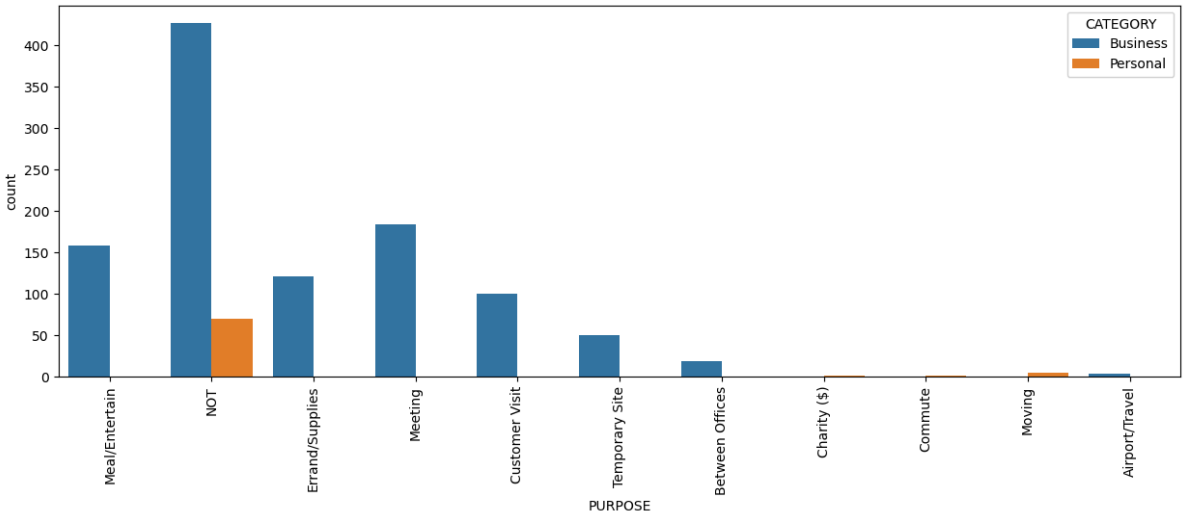
```
sns.countplot(dataset['day-night'])
plt.xticks(rotation=90)
```

```
C:\Users\yashc\anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarnin
g: Pass the following variable as a keyword arg: x. From version 0.12, the only va
lid positional argument will be `data`, and passing other arguments without an exp
licit keyword will result in an error or misinterpretation.
  warnings.warn(
```

Out[15]:
```
(array([0, 1, 2, 3]),
 [Text(0, 0, 'Morning'),
  Text(1, 0, 'Afternoon'),
  Text(2, 0, 'Evening'),
  Text(3, 0, 'Night')])
```



In [16]:
```
plt.figure(figsize=(15, 5))
sns.countplot(data=dataset, x='PURPOSE', hue='CATEGORY')
plt.xticks(rotation=90)
plt.show()
```



Insights from the above count-plots :

Most of the rides are booked for business purpose.

gs and Meal / Entertain purpose.

Most of the cabs are booked in the time duration of 10am-5pm (Afternoon).

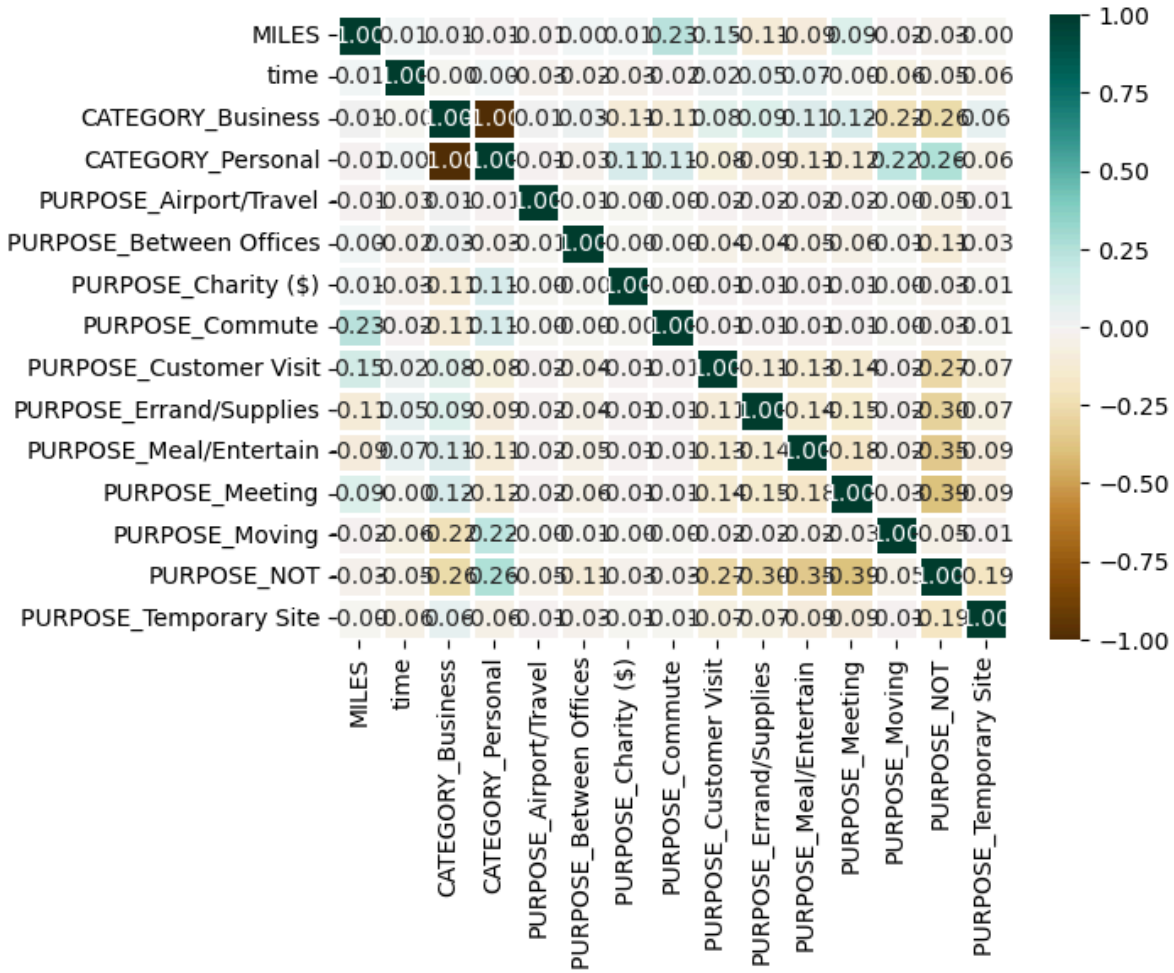
In [17]:
```
from sklearn.preprocessing import OneHotEncoder
object_cols = ['CATEGORY', 'PURPOSE']
OH_encoder = OneHotEncoder(sparse=False, handle_unknown='ignore')
OH_cols = pd.DataFrame(OH_encoder.fit_transform(dataset[object_cols]))
OH_cols.index = dataset.index
OH_cols.columns = OH_encoder.get_feature_names_out()
df_final = dataset.drop(object_cols, axis=1)
dataset = pd.concat([df_final, OH_cols], axis=1)
```

In [18]:
```
# Select only numerical columns for correlation calculation
numeric_dataset = dataset.select_dtypes(include=['number'])

sns.heatmap(numeric_dataset.corr(),
            cmap='BrBG',
            fmt='.2f',
            linewidths=2,
            annot=True)

# This code is modified by Susobhan Akhuli
```

Out[18]: <AxesSubplot:>



In [19]:
```
dataset['MONTH'] = pd.DatetimeIndex(dataset['START_DATE']).month
month_label = {1.0: 'Jan', 2.0: 'Feb', 3.0: 'Mar', 4.0: 'April',
               5.0: 'May', 6.0: 'June', 7.0: 'July', 8.0: 'Aug',
               9.0: 'Sep', 10.0: 'Oct', 11.0: 'Nov', 12.0: 'Dec'}
dataset["MONTH"] = dataset.MONTH.map(month_label)
```

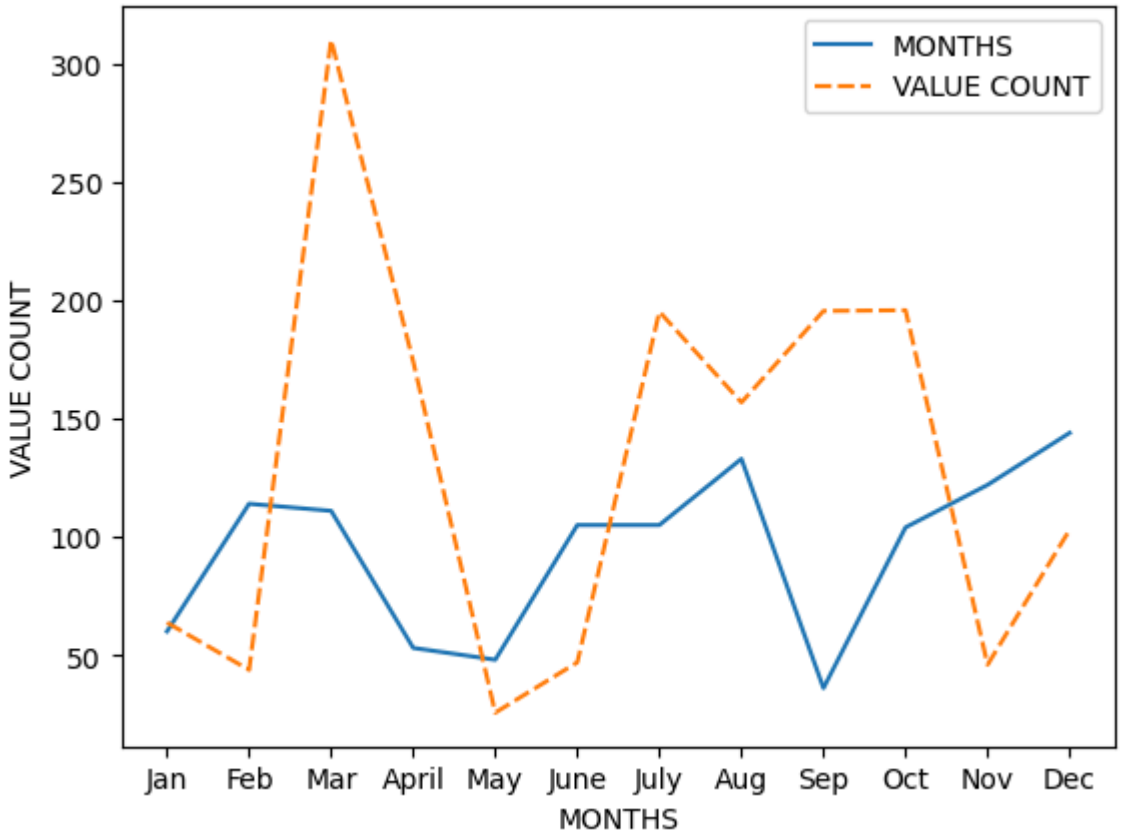


```
t=False)
```

```
# Month total rides count vs Month ride max count
df = pd.DataFrame({"MONTHS": mon.values,
                   "VALUE COUNT": dataset.groupby('MONTH',
                                                  sort=False)['MILES'].max()})

p = sns.lineplot(data=df)
p.set(xlabel="MONTHS", ylabel="VALUE COUNT")
```

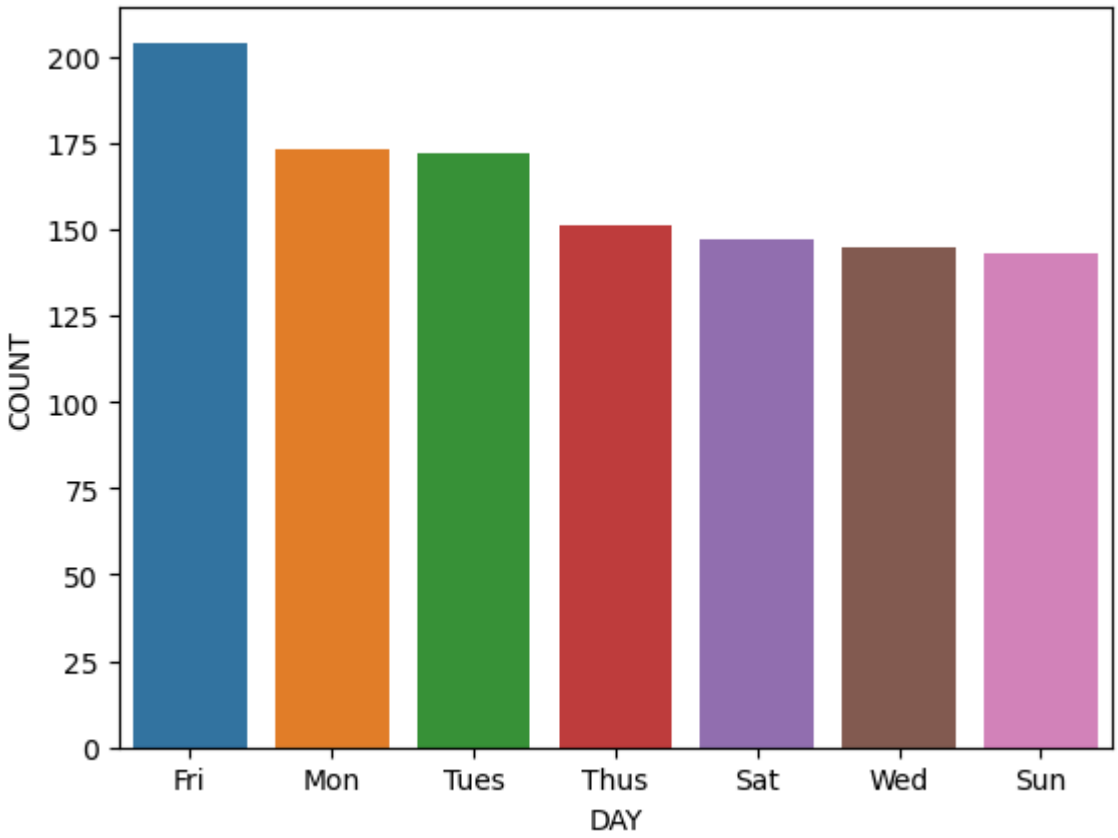
Out[19]:
```
[Text(0.5, 0, 'MONTHS'), Text(0, 0.5, 'VALUE COUNT')]
```



In [20]:
```
dataset['DAY'] = dataset.START_DATE.dt.weekday
day_label = {
    0: 'Mon', 1: 'Tues', 2: 'Wed', 3: 'Thus', 4: 'Fri', 5: 'Sat', 6: 'Sun'
}
dataset['DAY'] = dataset['DAY'].map(day_label)
```

In [21]:
```
day_label = dataset.DAY.value_counts()
sns.barplot(x=day_label.index, y=day_label);
plt.xlabel('DAY')
plt.ylabel('COUNT')
```

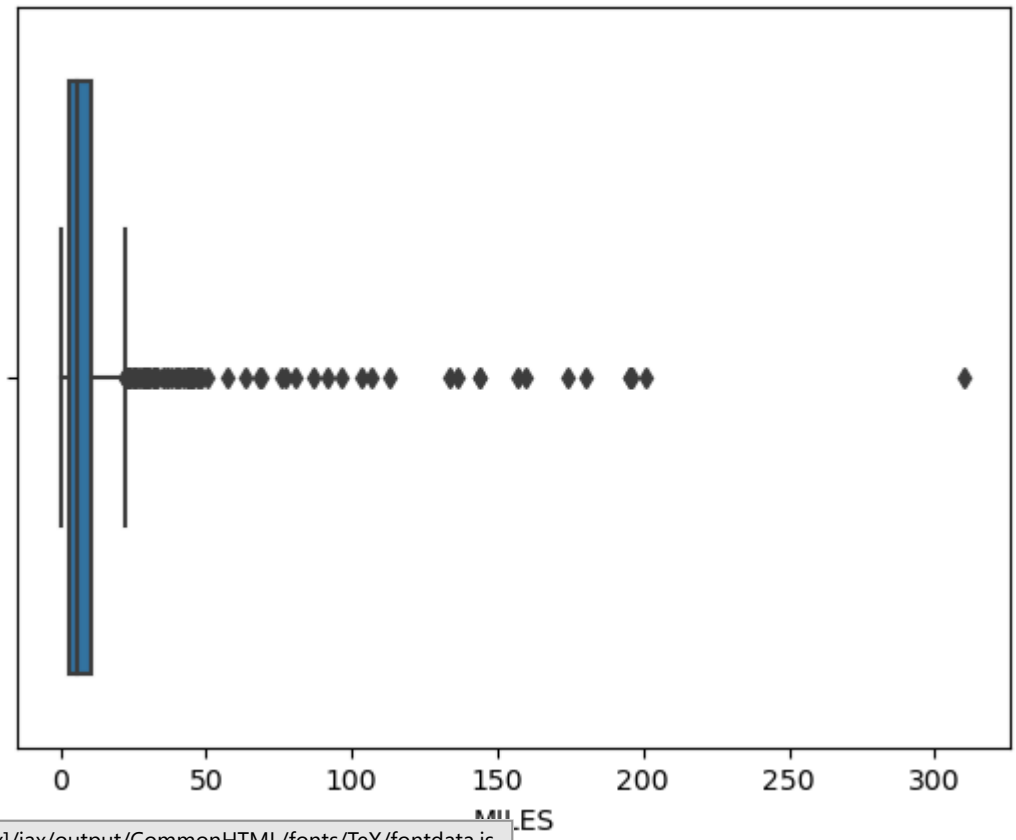
Out[21]:
```
Text(0, 0.5, 'COUNT')
```



In [22]:
```
sns.boxplot(dataset['MILES'])
```

```
C:\Users\yashc\anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarnin
g: Pass the following variable as a keyword arg: x. From version 0.12, the only va
lid positional argument will be `data`, and passing other arguments without an exp
licit keyword will result in an error or misinterpretation.
  warnings.warn(
```

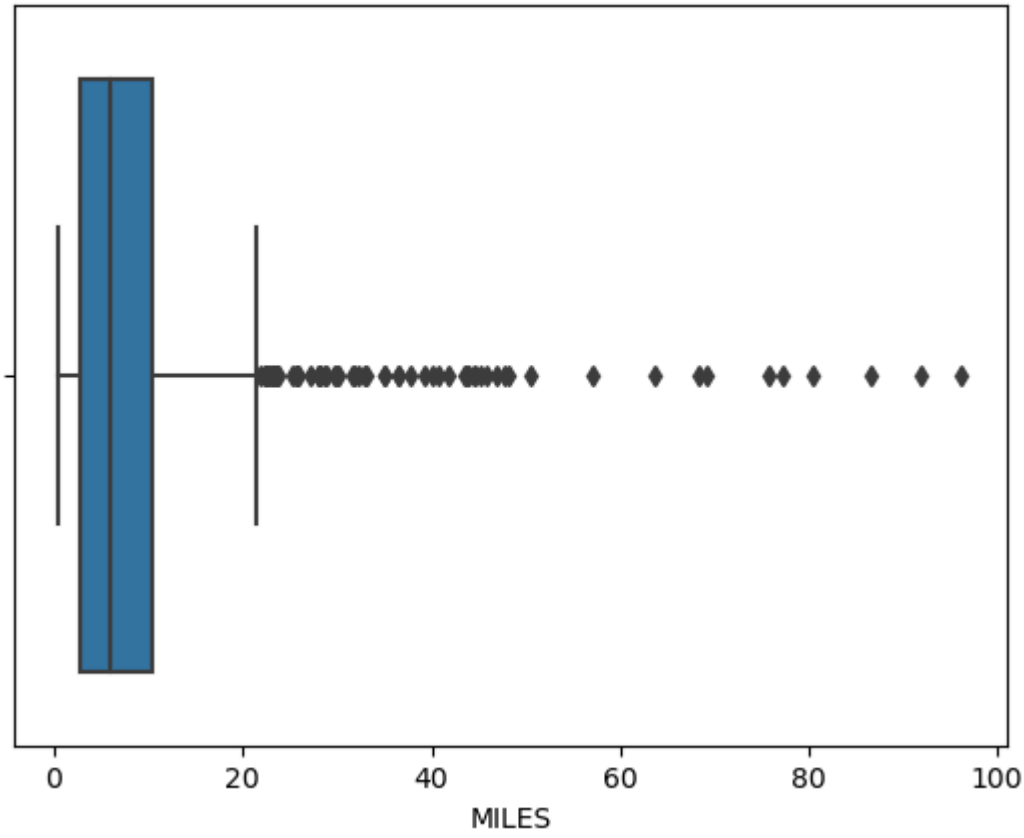
Out[22]:
```
<AxesSubplot:xlabel='MILES'>
```



In [23]:
```
sns.boxplot(dataset[dataset['MILES']<100]['MILES'])
```

```
C:\Users\yashc\anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarnin
g: Pass the following variable as a keyword arg: x. From version 0.12, the only va
lid positional argument will be `data`, and passing other arguments without an exp
licit keyword will result in an error or misinterpretation.
  warnings.warn(
```

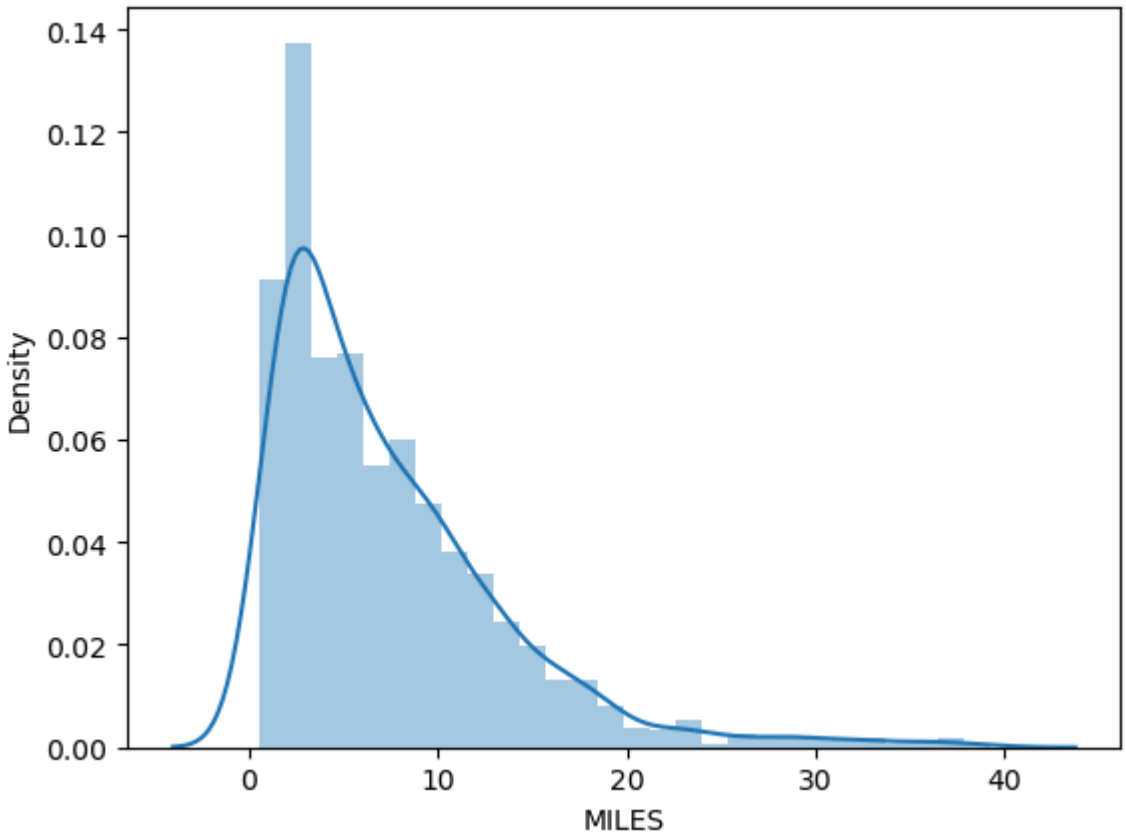
Out[23]:
```
<AxesSubplot:xlabel='MILES'>
```



In [24]:
```
sns.distplot(dataset[dataset['MILES']<40]['MILES'])
```

```
C:\Users\yashc\anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWa
rning: `distplot` is a deprecated function and will be removed in a future versio
n. Please adapt your code to use either `displot` (a figure-level function with si
milar flexibility) or `histplot` (an axes-level function for histograms).
  warnings.warn(msg, FutureWarning)
```

Out[24]:
```
<AxesSubplot:xlabel='MILES', ylabel='Density'>
```



In [ ]: